

RITIKA CHATTERJEE

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EDUCATION

Columbia University

Master of Science in Data Science, GPA: 3.6/4

Coursework: *Applied Deep Learning, Statistical Inference and Mathematical Modelling, Probability and Statistics*

New York, NY

Sep 2025 – Dec 2026

Vishwakarma Institute of Information Technology

Bachelor of Technology in Artificial Intelligence and Data Science, GPA: 9.5/10

Coursework: *Machine Learning, Artificial Intelligence, Computational Theory, Design and Analysis of Algorithms*

Pune, IN

Aug 2021 – May 2025

EXPERIENCE

Moby Filters Inc.

AI System and Automation Intern

- Configured and deployed a local AI workstation on Ubuntu to host and run multiple AI automation agents, built using LangGraph, and Python.
- Engineered a real-time IoT data pipeline for a filtration system using Raspberry Pi edge nodes, MQTT, and SQLite, with live sensor monitoring across flow meters and pressure sensors and automated fault detection to flag anomalies during filtration cycles.

New York, NY

May 2026 – July 2026

Columbia University

Research Assistant

- Built end-to-end ETL pipeline in Python to extract, clean, and standardize 50+ unstructured policy documents into a structured 54-feature dataset; designed schema, validation rules, and synonym mappings to ensure data quality and reproducibility.
- Conducted exploratory data analysis and applied unsupervised learning (k-means clustering, feature importance) to identify coverage-pattern segments across universities; delivered statistical findings, visualizations, and a documented codebase supporting a planned publication.

New York, NY

April 2026 – Present

Atlas Copco

Data Science Intern

- Built agentic AI systems using LLMs and RAG to automate enterprise data workflows, including a natural language-to-SQL agent (GraphRAG) that enabled non-technical users to query multi-database systems without writing SQL.
- Developed an AI-powered documentation and analytics agent for PowerBI for a NASA client, generating structured documentation and enabling natural language querying of metrics and data models, reducing manual effort by 40% and improving query turnaround by 30%.

Pune, IN

Jan 2025 – Jul 2025

NeuDay AI

AI Developer Intern

- Developed an AI-based exam evaluation system using OCR and LLMs, automating ~90% of grading; focused on aligning model outputs with grading rubrics and improving consistency, achieving a 35% improvement over manual evaluation.
- Designed a pipeline for real-time evaluation of scanned answer sheets, analyzing system performance across 100+ assessments and reducing instructor workload by 50% while providing structured feedback and analytics.

Pune, IN

Aug 2024 – Dec 2024

PUBLICATION & AWARDS

Decision Support System for Predicting Diseases using AI - IJARIE, 2024

Built a multi-disease ML prognosis system (Diabetes, Heart Disease, Parkinson's) using SVM, Random Forest, and Logistic Regression, with a Streamlit-based interface for real-time clinical decision support.

Tamer Social Enterprise Fellow Award, Columbia Business School, 2026

PROJECTS

Word Embedding Robustness Evaluation Framework

- Developed a comprehensive Python framework to evaluate the robustness of word embedding methods (Word2Vec, FastText, GloVe, BERT) against three data poisoning attack strategies, implementing automated training pipelines with PyTorch/TensorFlow and multiple evaluation metrics.

Dec 2025 – Feb 2026

Global Film Trends and Audience Engagement Analysis (IMDb Data)

- Analyzed IMDb datasets in R/Quarto to uncover global movie trends, including rating changes over decades, COVID-era audience engagement shifts, genre popularity, and increased cross-country release reach driven by streaming. Built interactive visualizations and a multi-page report.

Nov 2025 – Dec 2025

AI-Powered DSS for Ganges River Pollution level forecasting

- Led a 4-member team to build and deploy a real-time forecasting system predicting pollution levels 5 days ahead with alerts, collaborating with stakeholders for ongoing implementation; winner of the Smart India Hackathon 2024

Aug 2024 – Dec 2024

Vizathon 2.0

- Winner of Vizathon 2.0, a nationwide data visualization hackathon judged by Wolters Kluwer, recognized for innovative AI-driven insights by creating a real time analysis dashboard of retail market sales.

Nov 2023 – Dec 2023

TECHNICAL SKILLS

Language: Python, R, C++, SQL | **Technologies:** TensorFlow, PyTorch, Scikit-Learn, LangChain, LangGraph, Azure (OpenAI, Blob Storage, Resource Groups), Databricks, PowerBI Service, Docker, Streamlit, MySQL | **AI/LLM Systems:** LangChain, LangGraph, RAG, GraphRAG, Prompt Engineering, Agentic AI